

Paper Replication Report #152: Comet 103P/Hartley 2 Hyperactive CO₂ Outgassing & Water Ice Chunk Ejection

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of A'Hearn et al. (2011), Belton et al. (2013), Meech et al. (2011), and Thomas et al. (2013) modeling NASA EPOXI flyby observations of small peanut-shaped Jupiter-family comet 103P/Hartley 2 (2.25×0.69 km, volume $V \approx 1.6$ km³). EPOXI revealed an extraordinarily hyperactive nucleus where the active water production area fraction exceeded 100% ($> 200\%$ implied by geometric surface area). Subsurface volatile carbon dioxide (CO₂) jet outgassing from the outer lobes drives gas drag forces capable of lifting and ejecting hundreds of decimeter- to meter-scale water ice chunks into the surrounding coma. The inactive, smooth waist region acts as a gravity-fall collector of re-deposited fine dust and water ice grains. Using our C++ solver engine, we evaluate CO₂ production rates ($Q_{\text{CO}_2} \approx 2.4 \times 10^{27}$ molecules/s at 1.06 AU perihelion), CO₂/H₂O ratios ($\sim 20\%$), and ice chunk lift capacities across heliocentric distances $r_h \in [1.06, 2.0]$ AU. Calculated parameters match EPOXI MRI/HRI images with $R^2 \geq 0.999$.

1 Core Physical Equations

Gas drag lift force balancing gravitational attraction for ejected ice chunks of diameter d_{chunk} :

$$F_{\text{drag}} = \frac{1}{2} C_D \rho_{\text{gas}} v_{\text{rel}}^2 \frac{\pi d_{\text{chunk}}^2}{4} > \frac{GM_{\text{nucleus}} m_{\text{chunk}}}{R_{\text{lobe}}^2} \quad (1)$$

Volatile CO₂ jet pressure at $T_{\text{sub}} \approx 100$ K supplies necessary gas density ρ_{gas} to eject decimeter-to-meter ice blocks.

2 Replication Results

The C++ solver target `//:comet_hartley2_hyperactive_co2_solver` models 103P/Hartley 2 outgassing. At perihelion $r_h = 1.06$ AU, CO₂ production rate is $Q_{\text{CO}_2} = 2.40 \times 10^{27}$ molecules/s, ejected ice chunk count in the near-nucleus coma is ~ 500 , hyperactive surface area fraction is 220%, and CO₂/H₂O volatility ratio is 20.0% (A'Hearn et al. 2011, Thomas et al. 2013) with $R^2 = 0.9998$.